Mock Interview Edition

ETA Prediction: Correcting a Routing Engine in Milliseconds

18 scenarios with their follow-up probes and no answers. Use it to rehearse aloud, with a partner, or against a timer, then check yourself against the Study Edition.

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HOW THIS EDITION IS ORGANIZED

Section A, round plans: 5 ready-made sessions of five scenarios each, by theme. **Section B, question sheets:** every scenario followed by all of its probes, in order, each with its own notes box. The Study Edition page that holds the answers is printed under each scenario. **Section C, score sheet:** one row per scenario to track attempts. Question numbers and part numbers are identical in the Study Edition.

THREE WAYS TO USE IT

1. With a partner. The partner reads the scenario aloud, then asks the probes one at a time, moving on only after you have answered. **2. Alone, out loud.** Cover everything below the current probe with a sheet of paper, answer aloud for about two minutes, then slide the paper down to the next probe. **3. Alone, in writing.** Jot your key points in the notes box under each probe before looking at the next one, then compare with the Study Edition page printed under the scenario.

HOW TO GRADE A PROBE

Solid: you stated the main point, gave a number or a concrete example where one was possible, and said how you would check it. **Partial:** right direction but vague, or a number without the reasoning. **Missed:** you did not reach it, or reached the wrong conclusion. The last probe of every scenario asks for the general lesson in one sentence. Suggested pacing is about 12 minutes per scenario. Reading the answer only after you have committed to your own is what makes the practice work.

A NOTE ON THE PACING, GRADING AND ROUNDS

The 12-minute pacing, the three-level grading and the round compositions are the author's suggestions for practice, not an industry standard. The scenarios are written for this document and are not taken from a real company question bank. Where a scenario quotes a figure, the Study Edition says where it comes from.

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SECTION A

Round plans

Each round has five scenarios at about 12 minutes each, a 60-minute session. Mix and match freely. Click any scenario to jump to its question sheet; the number on the right is the page.

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SECTION B

Question sheets

Each scenario is followed by its probes in order. Indentation shows how deep the follow-up goes.

PART 1

Framing the problem: post-processing, different consumers and the sizing

Why correct a routing engine instead of replacing it, what different consumers need from an estimate, and where a 3 millisecond model spends its time.

01 Correct the routing engine, or replace it?

FRAMING

SCENARIO

A new engineer proposes dropping the routing engine ETA and training one model to predict the arrival time from trip features alone. The paper's design predicts a residual on top of the routing engine ETA. Argue both sides, and say what the sources prove.

OPENING PROBE

What does the routing engine do well?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What can it not know?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What practical reasons does the blog give for post-processing?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What do the sources prove?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 6

02 Which error matters to whom?

OBJECTIVE

SCENARIO

The fares team wants the best mean error. The delivery team wants the 95th percentile kept down. A manager asks for one model that does both. Explain what the paper does, using its Table 3.

OPENING PROBE

What do the consumers need?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does the paper do about it?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does Table 3 show?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What would you tell the manager?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 7

03 How much traffic is there?

ESTIMATION

SCENARIO

The paper says Uber has billion-level requests a week and that this is the highest-QPS model at Uber. It also describes its training data. Size the traffic and the fleet, being clear which anchor is which.

OPENING PROBE

What does the training data give?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Is that consistent with billion-level a week?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

How many requests are in flight?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

How many hosts?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 8

04 Where do the 3.25 milliseconds go?

LATENCY

SCENARIO

The model has 141.2 million parameters and answers in a median 3.25 ms. A teammate says: the arithmetic of a 141-million-parameter model must be the cost. Check it.

OPENING PROBE

How many parameters are dense?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does the dense work cost?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Then what takes the time?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does the attention layer add?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 9

PART 2

Data and features: discretize, geohash and hash

Why a network does better on binned inputs, what a geohash resolution means for table size, and when hashing helps.

05 Why bin the continuous features?

FEATURES

SCENARIO

The paper bucketizes continuous features such as speed into 256 quantiles before embedding them, and says this beat using the raw value. It also says quantile buckets beat equal-width ones. A skeptic says a network can learn any function of a number, so binning only throws information away. Respond with a worked example.

OPENING PROBE

What does the paper report?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Bin ten speeds two ways.

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Compute the information.

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does binning cost?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 10

06 How big would the location table be?

GEOSPATIAL

SCENARIO

The model encodes origin and destination as geohashes at lengths 4, 5, 6 and 7. A teammate wants an exact embedding for every cell. Size it, and size the hashed alternative.

OPENING PROBE

How many cells?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What is an exact table at u = 7?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does hashing cost?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What is lost?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 11

07 Hashing helped here. Why not at Google?

DESIGN CHOICE

SCENARIO

The CTR paper in this series reports that Google tried feature hashing and rejected it. Uber's paper uses hashing and says removing it hurts. A reviewer says the two results contradict each other. Resolve it.

OPENING PROBE

What does Uber's ablation say?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does the blog add?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Why might it differ from Google's result?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What would you do?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 12

08 Review the location hashing

CODE REVIEW

SCENARIO

A colleague writes this location feature. Offline accuracy is fine, then after deployment the live predictions are worse than offline, and two serving hosts disagree about the same request. Find the faults.

```
def location_bins(lat, lng, resolutions=(4, 5, 6, 7), bins=1_000_000):
    out = []
    for u in resolutions:
        cell = f"{round(lat, u)},{round(lng, u)}"
        out.append(hash(cell) % bins)
    return out
```

OPENING PROBE

Does it make geohash cells?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Is the hash stable?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What else is missing?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

State the fix.

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 13

PART 3

The model: linear attention, calibration, the loss and the results tables

Choosing an attention variant by what it costs, adjusting for request types, building a loss that fits the business, and reading the tables honestly.

09 Choose the attention variant

ARCHITECTURE

SCENARIO

The paper tried standard Transformer attention, found it too slow, and chose linear attention. A reviewer asks for the cost formulas, when the saving matters, and whether the claim is checkable. Answer.

OPENING PROBE

What are the costs?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

Compute the blog's example.

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

When does it matter?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

Is the paper's claim checkable?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What do you lose?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 14

10 One model for rides and deliveries

ARCHITECTURE

SCENARIO

Rides and deliveries have quite different distributions. The paper adds a bias per request type. A colleague proposes just adding the request type as another feature. Compare, with the paper's results.

OPENING PROBE

What is the layer?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Show the idea with numbers.

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What are the alternatives?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does the ablation show?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 15

11 Build the loss

LOSS

SCENARIO

A product owner says: being late costs more than being early, and delivery cares about the tail. Explain the paper's loss, compute it for an example, and say what the paper does not publish.

OPENING PROBE

What is the loss?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

Which direction is which?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

Compute with delta = 60 and omega = 0.3 (illustrative).

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What does that do to the prediction?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What is not published?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 16

12 Read the results tables

EVALUATION

SCENARIO

A slide says DeeprETA improves ETA error by 7.83% and also by 2.9%. A listener asks which is right. Use the tables to explain, and say what else the tables show.

OPENING PROBE

What does 7.83% measure?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does 2.9% measure?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What do the ablations in Table 2 show?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does the split by type show?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 17

13 Review the loss function

CODE REVIEW

SCENARIO

A colleague implements the asymmetric Huber loss for training. After training, the model's ETAs run too early and delivery customers complain that couriers are often late. The configured omega is 0.3. Find the fault.

```
def asym_huber(y, y_hat, delta=60.0, omega=0.3):
    e = abs(y - y_hat)
    base = 0.5 * e * e if e < delta else delta * e - 0.5 * delta * delta
    w = omega if y > y_hat else 1.0 - omega
    return w * base
```

OPENING PROBE

What should the weights be?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does the code do?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Price an error of 120 seconds.

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

State the fix, and tests.

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 18

PART 4

Serving, evaluation and operations

The request path, two figures that do not agree, comparing a model with an old one and with a base system, and watching for drift.

14 Trace one ETA request

SERVING

SCENARIO

A product manager asks what happens, and in what order, when an app shows an arrival time, and what the team needs to keep running. Trace it using the blog.

OPENING PROBE

What is the path?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What happens to the model's output?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What hardware?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does it depend on?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 19

15 Two sources, two figures

SOURCES

SCENARIO

The blog says any one prediction touches roughly 0.25% of the model's parameters. The paper's Table 4 says 0.5 million of 141.2 million parameters are outside the embeddings. A reviewer asks which is right. Work it out, and say what you would write.

OPENING PROBE

What does each say, in counts?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

Do they conflict?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What do the sources agree on?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What would you write?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What does the size mean for operations?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 20

16 Compare with the old model, and region by region

EVALUATION

SCENARIO

A VP asks whether the new model is better than the old one, and whether it is better everywhere. Answer with the shadow test and the city table, and say what the cities show.

OPENING PROBE

How was the new model compared with the old one?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What were the results?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Is it better everywhere?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What do the gaps show?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 21

17 Watch for drift

MONITORING

SCENARIO

The model is retrained weekly. The routing engine team plans to change how segment times are estimated. Describe what to monitor and how the two changes interact.

OPENING PROBE

Why does a routing engine change matter?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What signal does the paper give?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What else would you watch?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

How does weekly retraining help?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 22

18 Risks, and ten times the traffic

RISK

SCENARIO

A launch review asks what can go wrong with a post-processing ETA model and what grows if requests grow ten times. Use the sources and the earlier sizing.

OPENING PROBE

What can go wrong?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does ten times the requests do?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does it do to training?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What gets easier?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 23

SECTION C

Score sheet

Write how many probes were solid out of the total, per attempt. Revisit anything under half after reading the Study Edition's concepts box.

No.	Scenario	Probes	Attempt 1	Attempt 2	Attempt 3
01	Correct the routing engine, or replace it?	5			
02	Which error matters to whom?	5			
03	How much traffic is there?	5			
04	Where do the 3.25 milliseconds go?	5			
05	Why bin the continuous features?	5			
06	How big would the location table be?	5			
07	Hashing helped here. Why not at Google?	5			
08	Review the location hashing	5			
09	Choose the attention variant	6			
10	One model for rides and deliveries	5			
11	Build the loss	6			
12	Read the results tables	5			
13	Review the loss function	5			
14	Trace one ETA request	5			
15	Two sources, two figures	6			
16	Compare with the old model, and region by region	5			
17	Watch for drift	5			
18	Risks, and ten times the traffic	5			